TAFWELA

Key Performance Indicators (KPIs)

(DPEI) | April 2026

1. KPI Summary by Category

Category	# of KPIs	Primary Owner	Focus Area
Performance	5	Ziad, Mohannad	Speed & responsiveness
Reliability	4	Mohannad, Khaled	Stability & uptime
Usability	4	Youssef, Mohamed	User experience
Data Quality	3	Mohannad, Ziad	Accuracy & completeness
Development	4	Khaled	Code quality & process
Delivery	3	All Team	Milestones & documentation
TOTAL	**23**		

2. KPI Register

2.1 Performance KPIs

ID	KPI Name	Target	How to Measure	Tool	Owner	Status		
KPI-01	**Map Load Time**	< 2 seconds on 4G connection	Time from app launch to map fully rendered with markers	Flutter DevTools / Stopwatch	**Zaid**	Tracked		
KPI-02	**Congestion Status Update Latency**	< 1 second via Firestore listener	Time between staff update and user seeing the new status	Firestore Timestamps	Mohannad	Tracked		
KPI-03	**GPS Location Detection Time**	< 3 seconds on first launch	Time for Geolocator to return user coordinates	Flutter DevTools	**Zaid**	Tracked		
KPI-04	**App Startup Time (Cold Launch)**	< 3 seconds	Time from app icon tap to home screen ready	Flutter Performance Overlay	Khaled	Tracked		
KPI-05	**Station List Load Time**	< 1.5 seconds	Time to display list view with all nearby stations	Flutter DevTools	Khaled	Tracked		

2.2 Reliability KPIs

ID	KPI Name	Target	How to Measure	Tool	Owner	Status		
KPI-06	**App Crash Rate**	< 1% during testing phase	Number of crashes / total test sessions × 100	Flutter Crashlytics / Manual Log	Khaled	Tracked		
KPI-07	**Firebase Uptime**	> 99.5% during project period	Firebase Console uptime monitoring	Firebase Console Dashboard	Mohannad	Tracked		
KPI-08	**Real-time Sync Success Rate**	> 99% of status updates reflected correctly	Number of successful syncs / total updates × 100	Firestore Logs	Mohannad	Tracked		
KPI-09	**Navigation Button Success Rate**	> 98% successful Google Maps launches	Number of successful deep links / total taps × 100	Manual Testing Log	Mohannad	Tracked		

2.3 Usability KPIs

ID	KPI Name	Target	How to Measure	Tool	Owner	Status		
KPI-10	**GPS Permission Acceptance Rate**	> 90% of test users grant GPS permission	Number of users who accept GPS / total testers × 100	Manual Testing Log	**Zaid**	Tracked		
KPI-11	**Staff Status Update Success Rate**	> 99% of updates saved correctly to Firestore	Number of successful saves / total attempts × 100	Firestore Logs	**Omar**	Tracked		
KPI-12	**Search Response Time**	< 1 second to filter results	Time from search input to filtered list displayed	Flutter DevTools	**Mohamed**	Tracked		
KPI-13	**User Onboarding Completion Rate**	> 85% of testers complete onboarding flow	Number who reach home screen / total who open app × 100	Manual Testing Log	**Youssef**	Tracked		

2.4 Data Quality KPIs

ID	KPI Name	Target	How to Measure	Tool	Owner	Status		
KPI-14	**Seed Station Coverage**	Minimum 20 stations in Greater Cairo	Count of stations in Firestore with valid coordinates	Firestore Console	Mohannad	Tracked		
KPI-15	**GPS Accuracy**	Within 50 meters of actual user location	Compared against known GPS coordinates during testing	Manual Field Test	**Zaid**	Tracked		
KPI-16	**Data Consistency**	0 orphaned records in Firestore	Verify all station records have complete required fields	Firestore Console	Mohannad	Tracked		

2.5 Development KPIs

ID	KPI Name	Target	How to Measure	Tool	Owner	Status		
KPI-17	**Code Coverage (Unit Tests)**	> 60% of core business logic covered	flutter test --coverage report	Flutter Test + LCOV	Khaled	Tracked		
KPI-18	**GitHub Commit Frequency**	Minimum 3 commits/member/week	Count commits per member per week on GitHub	GitHub Insights	Khaled	Tracked		
KPI-19	**Pull Request Review Time**	< 24 hours per PR	Time from PR opened to merged/rejected	GitHub PR Logs	Khaled	Tracked		
			Time from bug					

2.6 Delivery KPIs

KPI-20	Bug Resolution Time	< 48 hours for High priority bugs	reported to fix merged	GitHub Issues	Khaled	Tracked

ID	KPI Name	Target	How to Measure	Tool	Owner	Status		
KPI-21	**Milestone Completion Rate**	100% of 4 milestones met on time	Milestones completed on schedule / total milestones × 100	Team Review Meetings	**team**	Tracked		
KPI-22	**APK Build Success**	1 stable APK delivered by May 12	Successful build with no critical crashes in first 10 min	Manual Testing	Khaled	Tracked		
KPI-23	**Documentation Completeness**	All 7 DPEI sections submitted	Count of submitted documents vs required	GitHub Repository	**team**	Tracked		

3. KPI Tracking Process

- KPIs reviewed at each weekly milestone meeting (every Sunday)
- Khaled maintains a shared KPI tracking sheet updated after each sprint
- Status updated from Tracked → Met or Not Met after measurement
- Any KPI marked Not Met triggers a team discussion for corrective action
- Final KPI report submitted as part of Section 7 (Final Presentation & Reports)